

Assignment 3: Inverse Differential Kinematics

16-384: Robot Kinematics and Dynamics

Prof. Jeff Ichnowski

Albert Xu

Henry Liao

Karthik Srinivasan

Due: See Canvas for the due date

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1 Overview

This assignment reinforces the following topics:

- Jacobians
- Psuedoinverses

2 Background

2.1 The Inverse Problem

So far we have studied the functions that perform forward kinematics, which means going from a manipulator's configuration space, Q to its workspace, W :

$$W = f(Q)$$

In other words, we have determined the function for a robot arm that uses its joint angles to calculate the position and orientation of its end effector (as well as other frames on the robot).

You can probably think of a lot of cases where we want to do the inverse: given a specified effector position (and potentially orientation), what are the joint angles that will achieve this? This inverse process is problematic because the forward kinematics function, f , is (usually) non-linear. That means that we (usually) cannot just say:

$$Q = f^{-1}(W)$$

This means that finding the inverse function for the kinematics of an arm is difficult (but by no means impossible—in the next assignment we will tackle several methods to do this).

2.2 Differential IK

While the general problem of going from workspace positions to joint angles is difficult, we can begin with the *differential* problem – going from workspace velocities to configuration space velocities.

$$\dot{q} = f^{-1}(\dot{w})$$

From the last assignment, you know that the function relating these two sets of velocities is the Jacobian, J . A key property of the Jacobian is that it is linear (given the configuration q of the arm). In practical terms, this means that if we know an arm's joint angles and desired instantaneous end-effector velocities, we can calculate the required joint velocities by inverting the Jacobian:

$$\dot{q} = J^{-1}(q)\dot{w}$$

Because (for a given configuration) the Jacobian can be represented as a matrix, this is a much easier problem than the general IK problem we began with. It doesn't directly give us the joint angles for a given end effector position, but it can be used to help solve for these, as we'll see in the assignment.

2.3 Pseudoinverse

But wait—although inverting matrices is easy for computers, not all matrices (and not all Jacobians) are invertible! In order for a matrix to be invertible, it has to be square, and have a non-zero determinant. In terms of a robot arm this means that it has to be:

- Fully constrained (the same number of workspace and configuration space variables)
- Non-singular (the arm's instantaneous motion can span the dimension of the workspace, meaning J is full rank)

Fortunately, mathematicians have developed a technique called the *pseudoinverse* to help get around the problem of not having a square or full rank matrix. The pseudoinverse has some, but not all, of the properties of a true matrix inverse. The general method to compute this quantity involves a technique called *singular value decomposition*, but when the matrix is *full rank*, there are analytical formulas that can be used to find the inverse.

In this case, we basically make the matrix square by multiplying it by its transpose. We then invert the new matrix, and multiply the result by the transpose of the original matrix to get the dimensions to match up. The exact order of multiplication depends on the dimension of the matrix:

2.3.1 Underconstrained

To determine the joint velocities that result in a given a workspace velocity, we need to solve a system of linear equations (given a joint configuration). If there are n joint variables and m workspace variables, then the Jacobian has dimensions $m \times n$. Consider the case where $m < n$ i.e. there are more joints than the number of workspace variables. This results in a system of m equations (i.e. constraints) in n unknowns (i.e. joint velocities) where $m < n$, hence the name underconstrained.

Assuming the Jacobian is full rank, such a system always has a solution. This means we can always find joint velocities that result in a given end effector velocity. However, there is a whole family of solutions that satisfy these constraints. Therefore, one might be interested in searching the best solution in this family, measured with respect to some criterion. For the purpose of this discussion, we are interested in solutions (joint velocities) which result in the smallest effort in moving the arm. We can quantify this effort using the squared norm of the joint velocities.

Posing this as an optimization problem, we want to find $\dot{q}$ such that we minimize $\dot{q}^T \dot{q}$ satisfying $\dot{w} = J(q)\dot{q}$, where $\dot{w}$ and $J(q)$ are known.

This optimization problem is equivalent to a least-norm problem for linear systems of equations and has an exact analytical solution given by:

$$\dot{q} = J^+ \dot{w},$$

where J^+ is known as the right pseudoinverse of the Jacobian and is given by:

$$J^+ = J^T (J J^T)^{-1}$$

2.3.2 Overconstrained

This case is when the number of rows the Jacobian has is greater than (or equal to) the number of columns. In other words, there are more equations than unknowns.

In this case, the arm cannot move the end effector with the exact desired velocity, but this solution will try to solve for the “best” joint velocities in terms of the sum-squared

error of end effector velocities. This is called the left pseudoinverse.

$$J^\# = (J^T J)^{-1} J^T$$

As a side note, the pseudoinverse in this overconstrained case is really useful even outside of kinematics, and is used to quickly and reliably do things like solve for best-fit lines to data.

2.3.3 Perfectly Constrained

Note that in the case of a square full rank matrix, the pseudoinverse will result in J^{-1} .

2.4 Course Logistics

In this class you would be using the following websites regularly:

- [Canvas](#): The main course web page. This links to everything relevant to the class (including the next two links).
- [Piazza](#): This is the best way to ask course staff questions, and see course announcements.
- [Gradescope](#): This is where you will be turning in your homework. You should have already been added to the course. If you haven't, please contact the course staff via email or Piazza post.

You can write your `writeup.pdf` in [Overleaf](#): upload the assignment zip as a new project, and use `assignment3.tex` as your template (it pulls in `16384_doc.cls` and the figures from the `latex/` folder automatically).

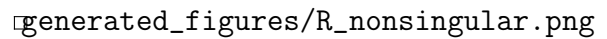
3 Instructions

- The deadline for this assignment is 17th September, 2026 11:00 A.M.
- The coding portion will be released by Friday this week, and will be due at the same time as this written portion. Necessary files will be made available on Github.
- The LaTeX format has been designed and tested on Overleaf, and **we strongly recommend using the Overleaf** to avoid any formatting errors as you fill out your answers.
- Zip your code into a single file named <AndrewId>.zip. See the complete submission checklist at the end, to ensure you have everything. Submit your PDF file to Gradescope.
- Each question (for points) is marked with a **points** heading.
- **Start early!** This homework may take a long time to complete.
- **During submission indicate the answer/page correspondence carefully when submitting on Gradescope.** If you skip a written question, just submit a blank page for it. This makes our work much easier to grade.
- If you have any questions or need clarifications, please post in Piazza or visit the TAs during the office hours.
- Unless otherwise specified, **all units are in radians, meters, and seconds**, where appropriate.

4 Written Questions

1) R Robot

Consider the following robot



Assume $\theta_1 = \frac{\pi}{4}$.

- a. [1 point] What is the dimension of the end-effector Jacobian? Consider end-effector positions $\begin{bmatrix} x \\ y \end{bmatrix}$ in $\mathbb{R}^2$.

- b. [1 point] Is this system underconstrained, overconstrained, or neither? Why?

- c. [2 points] Draw the velocity vector for the end effector caused by joint 1 moving with unit velocity. The direction of these vectors should be correct, but do not worry about the magnitude. You can complete this problem by inspection or by using properties of the Jacobian.

- d. [2 points] What is the dimension of the space spanned by this vector?

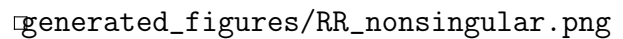
- e. [1 point] What is the rank of J for this configuration?

- f. [1 point] When identifying the inverse differential kinematics for this problem, would you use the inverse, the left psuedoinverse, or the right psuedoinverse?

- g. [3 points] What are the benefits of a left pseudoinverse versus a right pseudoinverse?

2) RR Robot

Consider the following robot

generated_figures/RR_nonsingular.png

Assume $\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \frac{\pi}{4} \\ \frac{\pi}{4} \end{bmatrix}$.

- a. [1 point] What is the dimension of the end-effector Jacobian? Consider end effector positions $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$.

- b. [1 point] Is this system underconstrained, overconstrained, or neither? Why?

- c. [4 points] On separate plots, draw the velocity vector for the end effector caused by:

- (i) Joint 1 moving with unit velocity (joint 2 has zero velocity).
- (ii) Joint 2 moving with unit velocity (joint 1 has zero velocity).

The direction of these vectors should be correct, and the relative magnitudes should be approximately correct, but do not worry about exact magnitudes. You can complete this problem by inspection or by using properties of the Jacobian.

- d. [2 points] What is the dimension of the space spanned by these vectors? Are the 1st and 2nd vectors linearly independent?

- e. [1 point] What is the rank of J for this configuration?

- f. [1 point] When identifying the inverse differential kinematics for this problem, would you use the inverse, the left psuedoinverse, or the right psuedoinverse?

3) RRR Robot

Consider the following robot

generated_figures/RRR_nonsingular.png

Assume $\begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \frac{\pi}{3} \\ \frac{\pi}{3} \\ \frac{\pi}{3} \end{bmatrix}$.

- a. [1 point] What is the dimension of the end-effector Jacobian? Consider end effector positions $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$.

- b. [1 point] Is this system underconstrained, overconstrained, or neither? Why?

- c. [6 points] On separate plots, draw the velocity vector for the end effector caused by
- (i) Joint 1 moving with unit velocity (joints 2 and 3 have zero velocity).
 - (ii) Joint 2 moving with unit velocity (joints 1 and 3 have zero velocity).
 - (iii) Joint 3 moving with unit velocity (joints 1 and 2 have zero velocity).

The direction of these vectors should be correct, and the relative magnitudes should be approximately correct, but do not worry about exact magnitudes. You can complete this problem by inspection or by using properties of the Jacobian.

- d. [2 points] What is the dimension of the space spanned by these vectors? Are the 1st and 2nd vectors linearly independent? The 1st and 3rd? The 2nd and 3rd? All three?

- e. [1 point] What is the rank of J for this configuration?

- f. [1 point] When identifying the inverse differential kinematics for this problem, would you use the inverse, the left psuedoinverse, or the right psuedoinverse?

4) RRR Robot, revisited with rotation
Consider the following robot

generated_figures/RRR_nonsingular.png

- a. [1 point] What is the dimension of the end-effector Jacobian, now considering $\begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$ end-effector positions in $SE(2)$.

- b. [1 point] Is this system underconstrained, overconstrained, or neither? Why?

- c. [1 point] Write out the end-effector Jacobian.

- d. [3 points] Assume $\theta_1 = 0.6$, $\theta_2 = 0.6$, and $\theta_3 = 0.4$ (all values in radians). Write out the end-effector velocity $\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$ for each of:

- (i) Joint 1 moving with unit velocity (joints 2 and 3 have zero velocity).
- (ii) Joint 2 moving with unit velocity (joints 1 and 3 have zero velocity).
- (iii) Joint 3 moving with unit velocity (joints 1 and 2 have zero velocity).

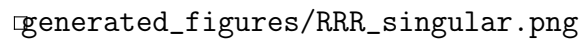
- e. [2 points] What is the dimension of the space spanned by these vectors? Are the 1st and 2nd vectors linearly independent? The 1st and 3rd? The 2nd and 3rd? All three?

- f. [1 point] What is the rank of J for this configuration?

- g. [2 points] When identifying the inverse differential kinematics for this problem, would you use the inverse, the left psuedoinverse, or the right psuedoinverse?

5) RRR Singular Configuration

Consider the following robot

generated_figures/RRR_singular.png

- a. [1 point] What is the dimension and rank of the end-effector Jacobian? Consider end effector positions $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$. Do the dimension and rank differ? why?

- b. [6 points] On separate plots, draw the velocity vector for the end effector caused by
- (i) Joint 1 moving with unit velocity (joints 2 and 3 have zero velocity).
 - (ii) Joint 2 moving with unit velocity (joints 1 and 3 have zero velocity).
 - (iii) Joint 3 moving with unit velocity (joints 1 and 2 have zero velocity).

The direction of these vectors should be correct, and the relative magnitudes should be approximately correct, but do not worry about exact magnitudes. You can complete this problem by inspection or by using properties of the Jacobian.

- c. [2 points] What is the dimension of the space spanned by these vectors? Are the 1st and 2nd vectors linearly independent? The 1st and 3rd? The 2nd and 3rd? All three?

- d. [1 point] What is the rank of J for this configuration?



6) Local Inverse Kinematics

Consider a general arm with revolute joints. Assume you already have the kinematic map to the end effector, f , and the Jacobian of the end effector, J .

The robot arm begins with joint angles Θ_0 , resulting in an end-effector position of $\mathbf{p}_0 = f(\Theta_0)$. The end effector needs to pick up an object at point $\mathbf{p}_1$.

Note: this is a conceptual problem: do not consider joint limits, and assume that all relevant positions are within the workspace of the robot. Furthermore, all answers are in terms of the given variables and known constants.

- a. [2 points] What are the initial joint angle velocities $\dot{\Theta}_0$ that start to move the end effector directly towards $\mathbf{p}_1$?

- b. [3 points] Assume the end effector has moved at $\dot{\Theta}_0$ for a small amount of time dt . What joint angles velocities $\dot{\Theta}_{dt}$ will move the end effector from this new starting point directly towards $\mathbf{p}_1$?

- c. [2 points] Describe (in a couple of sentences) how you would command the robot to move from $\mathbf{p}_0$ to $\mathbf{p}_1$ in approximately 1 second.

Note that this process describes how differential IK can be used to solve IK *numerically*, given a nearby starting point. We will explore this concept in more depth in the next assignment.

5 Code Questions

Ensure Python and the NumPy and Matplotlib libraries are correctly installed.
For Python installation, visit:

<https://www.python.org/downloads/>

For NumPy and Matplotlib, install via pip:

```
pip install numpy matplotlib
```

After setting up, download the code folder from the [GitHub repository](#) and navigate to that location. Whenever you work on the assignment, run your scripts from inside this directory.

1) [50 points] General Robot Class

Open directory code. For this exercise, we will walk you through the creation of a general Python class representing a fixed-base kinematic chain. You will implement forward kinematics for any RR arm. First, open the file `Robot.py`, and study this file. This is a basic Python class. The idea here is that you can pass parameters to the *constructor* of the robot class, and the returned object then has methods that can be called, and can operate on the values you passed in. For example, the following code:

```
robot = Robot(np.array([[0.55], [0.4]]), \
np.array([[1], [1]]), np.array([[1], [1]]), 0)

frames = robot.fk(np.array([[0], [np.pi/2]]))
```

can be used to find the forward kinematics of a robot with link lengths of 0.55 and 0.4 for $\theta = \begin{bmatrix} 0 \\ \frac{\pi}{2} \end{bmatrix}$



Forward Kinematics Your job is to write forward kinematics that can work for any number of revolute joints in a chain. You will fill in the `forward_kinematics` method in the `Robot` class. This computes H_i^0 for each frame ($i > 0$) in the figure above, including the end effector frame. The code describes the conventions used for the resulting variable. One suggestion that should help is to break this problem into three components:

- First consider frame 1.
- Use a for loop to iterate from frame 2 to $n - 1$; use H_{i-1}^0 to help compute frame H_i^0 .
- Compute the end effector frame, using H_{n-1}^0 to help.

Verification and Submission To verify your results:

1. Run the `sample_path` function. This will generate a plot and save two CSV files: `calculated_path.csv` and `ground_truth_path.csv`. This is for your own reference, to check whether your output matches the sample test case.
2. Run the local self-check before submitting: from the `local_autograder` directory, run

```
python local_autograder.py
```

This runs your `Robot.fk` over 25 published test arms (chain lengths 2 through 6) and prints PASS/FAIL for each. A FAIL message tells you what to look for:

- `<n>/<total> poses off by > 1 cm` — fk runs but returns the wrong position; check your homogeneous transforms.
 - `raised <ExceptionType>: <message>` — fk crashed; the message says where.
 - end-effector path has the wrong shape or is not finite — your output isn't the expected $(N, 2)$ array, or it contains NaN/inf.
3. The autograder on Gradescope will check your implementation against the same 25 arms and joint angles, but perturbs each one a little and checks against the reference solution instead of a file. If the local check above passes, you should pass on Gradescope too.
 4. To submit your work:
 - Run the `create_submission.py` script provided in the code directory. This will zip up all necessary files in the code folder.
 - Submit the resulting zip file to Gradescope.

Save your sample path plot as a PNG and include it in your writeup. Also include an explanation of your forward kinematics implementation and any challenges you encountered.

6 Hands-on Lab

So far the arms in this assignment have been drawings. In this section you will put your own kinematics on a real one: a seven-joint UFACTORY xArm7, driven as the planar RR arm you have been working with all along. You will implement its forward kinematics and its Jacobian, check them against the simulator from your own machine, and then — in lab — push the real arm by hand through a path of your choosing and watch it replay that path while your kinematics draw it live.

The robots live in the *Robotics Engineering Lab*. Please be mindful of others and share the robots fairly: check Piazza for signups.

6.1 The xArm7 as a Planar RR Arm

The xArm7 has seven revolute joints; the arm in the written questions has two, in a plane. The two become the same mechanism once four of the seven joints are held still. Holding

$$\theta_2 = \frac{\pi}{2}, \quad \theta_3 = \frac{\pi}{2}, \quad \theta_5 = -\frac{\pi}{2}, \quad \theta_6 = \frac{\pi}{2}$$

leaves joints 1 and 4 turning about axes that are both parallel to the world z -axis, so the end effector moves in a horizontal plane and the arm is an RR arm lying on its side. (Joint 7 is left free as well, but its axis is that same vertical one and it only spins the tool about itself, so it does not move the point you are tracing.) Joints 1 and 4 are the *free* joints, and their angles — after a fixed offset, which `robot_info.py` handles for you — are the θ_1 and θ_2 of your RR arm.

The two links are not round numbers, because they come from where the xArm7's joints actually are: $l_1 = 0.2977$ m from the shoulder to the elbow and $l_2 = 0.4256$ m from the elbow to the wrist. Read them from `robot_info()['link_lengths']` rather than typing them in — the starter code already does this for you.

6.2 Setting Up

The lab code needs the course robot library, `xarm7_lib`. If you installed it earlier in the semester, update it, since the version this assignment needs is newer:

```
git clone https://github.com/CMU-16384-2026-Fall/16384-robot-lib
cd 16384-robot-lib
git pull                                # if you already had it cloned
conda env update -f environment.yml
conda activate 16384
```

Then get this assignment's lab code, which is the `lab` folder of the same [GitHub repository](#) you downloaded code from:

```
cd Assignment3/lab
```

Run every command in this section from inside that lab folder. Work in a Git repository that you can push from and pull onto the lab machine: you will write your code on your own machine, and the fastest way to get it onto the robot's computer is `git pull`.

The same scripts drive either the real arm or a MuJoCo simulation, and they choose between them by looking at the `ROBOT_IP` environment variable: set, and they talk to the arm at that address; unset, and you get the simulation. So everything below except the recording itself can be developed and debugged at home, with no robot.

1) [50 points] Tracing a Path on the xArm7

This question is graded by hand, from one video you record in lab. Read the deliverable at the end before you go, so you know what the video has to show.

To-Do 1: Forward Kinematics Open `fk.py` and fill in `forward_kinematics_RR`, which returns two 3×3 planar homogeneous transforms. Number the frames along the chain the way you did in the last assignment — a rotation at each joint, a translation along each link — so that $\mathbf{H}_2^0$ is the frame at the far end of link 1 (the elbow) and $\mathbf{H}_4^0$ is the frame at the far end of link 2 (the end effector), both expressed in the base frame.

To-Do 2: The Jacobian Fill in `jacobian_RR`, the 2×2 end-effector *position* Jacobian $\mathbf{J}$ at (θ_1, θ_2) : column i is how fast the end effector moves, and in what direction, when joint i turns at unit speed. It is what turns joint velocities into the end effector's velocity,

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \mathbf{J}(\theta_1, \theta_2) \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix},$$

and the plot below draws exactly that product as an arrow off the end effector, so a correct Jacobian is something you can see.

Checking Your Work in Simulation The lab folder ships a sample recording, `square.csv`, so you can test both functions before you ever touch a robot. With `ROBOT_IP` unset, run:

```
python replay.py square.csv
```

Two windows open: the simulated xArm7 in a browser tab (its address is also printed as `Meshcat: ...` when the script starts), and a top-down plot of the arm drawn entirely by *your* code. The plot is drawn as if you were standing in front of the robot, looking down: the base is at the top, the robot's $+x$ axis points down the screen towards you, and its $+y$ axis to your right. In it you should see

- the blue stick figure lying on top of the simulated arm, with its elbow at the simulated arm's elbow;
- the solid black path the end effector traces landing on the dotted orange path it was asked to follow;

- the green arrow pointing along the direction of travel, growing when the arm moves faster and shrinking to nothing when it stops.

If something is off, the symptom tells you which To-Do to look at:

- The drawn arm does not match the simulated arm — `forward_kinematics_RR`. Note that only the translation column of each transform is used to draw the arm, so check the rotation blocks too; they can be wrong while the picture still looks reasonable.
- The path is right but the arrow points the wrong way, or has wildly the wrong length — `jacobian_RR`.
- a 0.213 rad jump at sample 57 — the samples reaching the arm are out of order, in degrees rather than radians, or θ_1 and θ_2 the wrong way round.

Sort all of this out before your lab session; the arm is a slow place to debug a sign error.

Safety: Read this before you touch the robot

- **Powering on.** Ask a TA during their office hours to walk you through powering up and enabling your station the first time. Do not power-cycle the controller yourself if the arm is in an unexpected state; ask.
- **The e-stop.** Know where the emergency stop button is *before* you start, and keep it within reach. Press it for anything you did not expect — an unexpected motion, a noise, a cable catching. Stopping the arm for no reason costs you nothing.
- **Stay out of the workspace.** Never put any part of yourself inside the arm's reach while it can move under its own power. The only time you touch the arm is when you are hand-guiding it, and the scripts tell you when that is.
- **“Press enter when your hands are clear” means it.** The recording pauses with that prompt whenever it is about to move the arm on its own. Let go, step back, then press enter.
- **The safety box.** The arm is confined to a box around its base, drawn in red on the live plot:

$$x \in [-0.10, 0.70] \text{ m}, \quad y \in [-0.45, 0.45] \text{ m}, \quad z \in [0.00, 0.80] \text{ m}.$$

A command that would take the arm out of the box is refused outright rather than started and stopped part-way. The drawn box is accurate, but the drawn *arm* is a stick figure and the real one has width, so the real arm will trip the boundary before the drawing appears to touch it.

- **The motors stay on.** When a script exits, the arm is left stopped but still energized and holding its position. That is deliberate: switching the motors off would let it fall under its own weight.

Recording a Path At your station:

1. Get your code onto the machine with `git pull`. You can edit in place with `vim` or `nano`, but it is usually easier to edit on your laptop, push, and pull again.
2. Start the recording, naming the file you want:

```
python record.py my-path.csv
```

The script first squares the locked joints (2, 3, 5 and 6) so that the arm is in the RR plane. If that move is refused because the arm is parked somewhere it is not allowed to move from, see **When the arm gets stuck** below, then run it again.

3. The arm then unlocks and you push it by hand. Trace a path you will recognize later — a square, a circle, your initials. Watch the live plot as you go: the blue arm should follow your hand, and the green arrow should point the way you are pushing.
4. Press Ctrl-C to stop. The recording is saved as a plain CSV of the two joint angles, sampled at 100 Hz; with no filename given it lands in `recordings/` with a timestamp for a name.

Replaying It Play the recording back on the arm with the same script you tested in simulation:

```
python replay.py my-path.csv
```

The arm moves to the first sample of the recording, then streams the rest of it. On the plot, the dotted orange path is what you recorded, and the solid black path is where your forward kinematics says the arm actually went — the arm's own measured joint angles, through your code.

When the Arm Gets Stuck Two things will happen to you, and neither is a breakage:

- **A locked joint drifts.** While you are hand-guiding, every joint is free, including the four that are supposed to stay put. Push the arm out of its plane and one of them will drift far enough that the recording stops itself, tells you to take your hands off, and — once you press enter — slowly puts that joint back before carrying on. Follow the prompts; it fixes itself.
- **The arm is parked somewhere it cannot move from.** If a script refuses to move the arm at all, hand-guide it back to a legal pose:

```
python -m xarm7_lib.free_drive
```

This releases every joint so you can push the arm wherever you want, and locks it there when you press enter. Then re-run the script that refused.

Deliverable Record **one video** of your replay, with both of these in frame at the same time:

- the real arm replaying the path you traced, with the whole arm visible; and
- the monitor showing the live plot — your arm drawn by your forward kinematics, the path it traces, and the green velocity arrow from your Jacobian.

A phone video is fine, in any common format (.mp4, .mov, .webm). Show at least one complete traversal of your path. Put the video, together with the recording it replays, in the lab/handin folder:

```
lab/handin/my-video.mp4
lab/handin/my-path.csv
```

create_submission.py picks up everything in that folder, along with your lab/fk.py, and puts it in the same zip as your Robot.py.

7 Submission Checklist

- ☐ Create a PDF of your answers to the written questions with the name `writeup.pdf`.
- ☐ Make sure `writeup.pdf` includes your saved sample-path plot and FK write-up from the Code Questions section.
- ☐ Complete `forward_kinematics_RR` and `jacobian_RR` in `lab/fk.py`.
- ☐ Confirm `python replay.py square.csv` looks right in simulation before your lab session.
- ☐ Record a path on the real arm with `record.py`, and replay it with `replay.py`.
- ☐ Put your replay video and the recording it replays in the `lab/handin` folder.
- ☐ Run `create_submission.py` inside the code directory to generate the submission zip file.
- ☐ Upload the resulting `<andrew_id>_hw3.zip` to Gradescope.
- ☐ Upload `writeup.pdf` to Gradescope.